SHUBHAM KISHOR KALE

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AI/Data Engineer with ~2 years building large-scale ELT pipelines and agentic RAG/LLM systems across healthcare, finance, and insurance domains.

EDUCATION

Stony Brook University | New York, USA — **Aug 2024 - Dec 2025**
Master of Science in Data Science — **GPA: 3.78/4**

NMIMS University | Mumbai, India — **Jul 2019 - May 2023**
Bachelor of Technology in Data Science — **CGPA: 3.56/4**

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, SQL, Scala, JavaScript, TypeScript, Bash

Concepts & Domains: AI Engineering, Data Engineering, Data Science, Data Analysis, Cloud Computing, Machine Learning, Deep Learning, NLP, Generative AI (RAG, LLM Agents, MCP, Fine-tuning), Computer Vision, Statistical Analysis, Causal Inference, A/B Testing, Time Series Analysis, Data Structures & Algorithms, System Design, REST APIs, Server-Sent Events (SSE), OOP & Design Patterns, ETL, Data Ingestion, Data Warehousing, Data Modeling, Data Architecture, Data Security, Distributed Systems, Finance, Healthcare

Frameworks, Tools & Cloud: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, LangChain, LangGraph, Pandas, NumPy, Spark, Kafka, Docker, Kubernetes, Apache Airflow, Dagster, dbt, AWS (S3, Redshift, IAM), Azure, GCP (BigQuery), Hadoop, Snowflake, PostgreSQL, MongoDB, FastAPI, Flask, Streamlit, React, Next.js, Pydantic, pytest, Git, GitHub Actions (CI/CD), Tableau, Power BI

WORK EXPERIENCE

Quantegy Analytics | New York, USA — **May 2025 - Present**
Analytics Engineer

- Delivered production data platforms and BI across healthcare, insurance, and consumer-products clients for cross-functional and C-suite stakeholders.
- Engineered Dagster-orchestrated ELT pipelines ingesting 4M+ patient and financial records from 15+ Redshift tables into Snowflake, powering downstream analytics across 3 healthcare markets.
- Resolved extraction bottleneck via parallel multi-table ingestion (~5x throughput) with REST API integration, S3 archival, and Pydantic schema validation in a Dockerized CLI framework adopted by 4+ teams.
- Productionized scheduling on Airflow with full pytest/httpx coverage validating 10M+ records across 10+ tables.
- Implemented data security and governance via AWS IAM/VPN access policies, Secrets Manager credentials, role-based Snowflake access across 2 client environments, and PII redaction for credit-card data.
- Shipped a Streamlit analytics platform used daily by 50+ users across 5 markets (financial, patient-volume, and case-tracking dashboards), compressing a 3-day Excel reporting cycle to same-day.

Mettler-Toledo | Mumbai, India — **Mar 2024 - Jun 2024**
Data Analyst Intern

- Designed an end-to-end RAG system with fine-tuned T5/BART models (Hugging Face) for automated KPI commentary, with vector search over 10M+ financial records (85%+ relevance); removed 15 hrs/month of manual write-ups.
- Automated credit-representative assignment via a rule-based engine, cutting allocation time 50% and saving 30 hrs/month of operational overhead.

Marcellus Investment Managers | Mumbai, India — **Mar 2023 - Nov 2023**
Quantitative Analyst Intern

- Consolidated 25 tables (3M+ rows) across 2 PostgreSQL schemas (ACE Equity, Bloomberg) and created Python automation that cut 10 hrs/week of manual reconciliation.
- Developed a regression-based impact-cost model ($R^2 = 0.82$) for trade-execution optimization and a forensic-screening ML ensemble (84% accuracy, 87% specificity) adopted into ongoing fraud-risk screening.

PROJECTS

AI Financial Research Agent · [GitHub] · [Live Demo] · *Full-Stack GenAI, Agentic RAG, MCP, SSE Streaming, Evaluation*

- Architected an agentic RAG system over SEC 10-K filings (5 companies, 67K+ chunks) using LangChain, LangGraph ReAct agents, FastMCP server (streamable-HTTP Model Context Protocol), ChromaDB with MiniLM embeddings, and Gemini 2.5 Flash Lite; tools discovered at runtime via MCP, decoupling agent logic from tool implementations.
- Deployed as a two-service FastAPI + MCP microservice on Docker Compose with MCP-first execution, direct-agent fallback, and 30s timeout handling; CI/CD via GitHub Actions runs lint, dry-run eval, and pytest with no secrets required.
- Built a RAGAS evaluation harness with a separate judge model to avoid quota contention with the agent's generation model; scored 0.80 faithfulness and 0.80 context recall across question types.